

No.	Solution
5(a)	$I = nAqv = \frac{Q}{t}$ $Q = nAqvt$ $= (5.9 \times 10^{28}) \left(\frac{\pi}{4} (0.38 \times 10^{-3})^2 \right) (1.60 \times 10^{-19}) (7.2 \times 10^{-5}) (30 \times 60)$ $= 140 \text{ C}$
5(b)(i)1.	$2\pi f = 120\pi$ $f = 60 \text{ Hz}$
5(b)(i)2.	$V_{r.m.s.} = \frac{9.0}{\sqrt{2}} = 6.36 \approx 6.4 \text{ V}$
5(b)(ii)1.	$\text{Total resistance} = 12 + \frac{12 \times 6.0}{12 + 6.0} = 12 + 4.0 = 16 \text{ } \Omega$ $I_0 = \frac{V_0}{R} = \frac{9.0}{16} = 0.56 \text{ A}$
5(b)(ii)2.	$V_{r.m.s.} \text{ across } 6 \text{ } \Omega \text{ resistor} = \frac{4.0}{16} \times 6.36 = 1.59 \text{ V}$ $\langle P \rangle = \frac{V_{r.m.s.}^2}{R} = \frac{1.59^2}{6.0} = 0.42 \text{ W}$
6(c)	Magnetic flux density is defined as the force per unit length per unit current acting